

Wine Count Regression

Darwhin Gomez

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Wine Count Regression

Overview In this homework assignment, you will explore, analyze and model a data set containing information on approximately 12,000 commercially available wines. The variables are mostly related to the chemical properties of the wine being sold. The response variable is the number of sample cases of wine that were purchased by wine distribution companies after sampling a wine. These cases would be used to provide tasting samples to restaurants and wine stores around the United States. The more sample cases purchased, the more likely is a wine to be sold at a high end restaurant. A large wine manufacturer is studying the data in order to predict the number of wine cases ordered based upon the wine characteristics. If the wine manufacturer can predict the number of cases, then that manufacturer will be able to adjust their wine offering to maximize sales. Your objective is to build a count regression model to predict the number of cases of wine that will be sold given certain properties of the wine. HINT: Sometimes, the fact that a variable is missing is actually predictive of the target. You can only use the variables given to you (or variables that you derive from the variables provided).

Variable Name	Definition	Theoretical Effect
INDEX	Identification variable (do not use)	None
TARGET	Number of cases purchased	None
AcidIndex	Proprietary method of testing total acidity of wine using a weighted average	
Alcohol	Alcohol content	
Chlorides	Chloride content of wine	
CitricAcid	Citric acid content	
Density	Density of wine	
FixedAcidity	Fixed acidity of wine	
FreeSulfurDioxide	Sulfur dioxide content of wine	
LabelAppeal	Marketing score indicating label appeal. High: customers like it. Negative: customers dislike it.	Higher label appeal suggests higher sales.
ResidualSugar	Residual sugar of wine	
STARS	Wine rating by expert panel (1–4 stars)	Higher number of stars suggests higher sales.
Sulphates	Sulfate content of wine	
TotalSulfurDioxide	Total sulfur dioxide content of wine	
VolatileAcidity	Volatile acid content of wine	
pH	pH of wine	

Data Exploration

In this section, describe the structure of the wine training dataset, highlight important summary statistics, identify missing values, examine relationships with the target variable, and show key visualizations.

Data-set Overview

The training dataset contains **12,795 observations** and **14 predictor variables**, and the target variable (**TARGET**). The variables include chemical measurements such as acidity, sulfur content, pH levels, alcohol concentration, and marketing-related attributes like **LabelAppeal** and **STARS**. The evaluation set holds **3,335 observations**, **there are all numeric variables with** **LabelAppeal**, **AcidIndex**, **Stars** being of the **int** type the other 11 predictors are numeric type.

Summary Statistics

Provide mean, median, standard deviation, and distribution shape for key variables.

Train Summary

```
wine_data<- wine_data|>
  dplyr::select(-INDEX)
summary(wine_data)
```

TARGET	FixedAcidity	VolatileAcidity	CitricAcid
Min. :0.000	Min. : -18.100	Min. : -2.7900	Min. : -3.2400
1st Qu.:2.000	1st Qu.: 5.200	1st Qu.: 0.1300	1st Qu.: 0.0300
Median :3.000	Median : 6.900	Median : 0.2800	Median : 0.3100
Mean :3.029	Mean : 7.076	Mean : 0.3241	Mean : 0.3084
3rd Qu.:4.000	3rd Qu.: 9.500	3rd Qu.: 0.6400	3rd Qu.: 0.5800
Max. :8.000	Max. : 34.400	Max. : 3.6800	Max. : 3.8600
ResidualSugar	Chlorides	FreeSulfurDioxide	TotalSulfurDioxide
Min. : -127.800	Min. : -1.1710	Min. : -555.00	Min. : -823.0
1st Qu.: -2.000	1st Qu.: -0.0310	1st Qu.: 0.00	1st Qu.: 27.0
Median : 3.900	Median : 0.0460	Median : 30.00	Median : 123.0
Mean : 5.419	Mean : 0.0548	Mean : 30.85	Mean : 120.7
3rd Qu.: 15.900	3rd Qu.: 0.1530	3rd Qu.: 70.00	3rd Qu.: 208.0

Max. : 141.150	Max. : 1.3510	Max. : 623.00	Max. : 1057.0
NA's : 616	NA's : 638	NA's : 647	NA's : 682
Density	pH	Sulphates	Alcohol
Min. : 0.8881	Min. : 0.480	Min. : -3.1300	Min. : -4.70
1st Qu.: 0.9877	1st Qu.: 2.960	1st Qu.: 0.2800	1st Qu.: 9.00
Median : 0.9945	Median : 3.200	Median : 0.5000	Median : 10.40
Mean : 0.9942	Mean : 3.208	Mean : 0.5271	Mean : 10.49
3rd Qu.: 1.0005	3rd Qu.: 3.470	3rd Qu.: 0.8600	3rd Qu.: 12.40
Max. : 1.0992	Max. : 6.130	Max. : 4.2400	Max. : 26.50
	NA's : 395	NA's : 1210	NA's : 653
LabelAppeal	AcidIndex	STARS	
Min. : -2.000000	Min. : 4.000	Min. : 1.000	
1st Qu.: -1.000000	1st Qu.: 7.000	1st Qu.: 1.000	
Median : 0.000000	Median : 8.000	Median : 2.000	
Mean : -0.009066	Mean : 7.773	Mean : 2.042	
3rd Qu.: 1.000000	3rd Qu.: 8.000	3rd Qu.: 3.000	
Max. : 2.000000	Max. : 17.000	Max. : 4.000	
		NA's : 3359	

Test Summary

```
wine_test<- wine_test|>
  dplyr::select(-TARGET)
summary(wine_test)
```

IN	FixedAcidity	VolatileAcidity	CitricAcid
Min. : 3	Min. : -18.200	Min. : -2.8300	Min. : -3.1200
1st Qu.: 4018	1st Qu.: 5.200	1st Qu.: 0.0800	1st Qu.: 0.0000
Median : 7906	Median : 6.900	Median : 0.2800	Median : 0.3100
Mean : 8048	Mean : 6.864	Mean : 0.3103	Mean : 0.3124
3rd Qu.: 12061	3rd Qu.: 9.000	3rd Qu.: 0.6300	3rd Qu.: 0.6050
Max. : 16130	Max. : 33.500	Max. : 3.6100	Max. : 3.7600
ResidualSugar	Chlorides	FreeSulfurDioxide	TotalSulfurDioxide
Min. : -128.300	Min. : -1.15000	Min. : -563.00	Min. : -769.00
1st Qu.: -2.600	1st Qu.: 0.01600	1st Qu.: 3.00	1st Qu.: 27.25
Median : 3.600	Median : 0.04700	Median : 30.00	Median : 124.00
Mean : 5.319	Mean : 0.06143	Mean : 34.95	Mean : 123.41
3rd Qu.: 17.200	3rd Qu.: 0.17100	3rd Qu.: 79.25	3rd Qu.: 210.00
Max. : 145.400	Max. : 1.26300	Max. : 617.00	Max. : 1004.00
NA's : 168	NA's : 138	NA's : 152	NA's : 157

Density	pH	Sulphates	Alcohol
Min. :0.8898	Min. :0.600	Min. : -3.0700	Min. : -4.20
1st Qu.:0.9883	1st Qu.:2.980	1st Qu.: 0.3300	1st Qu.: 9.00
Median :0.9946	Median :3.210	Median : 0.5000	Median :10.40
Mean :0.9947	Mean :3.237	Mean : 0.5346	Mean :10.58
3rd Qu.:1.0005	3rd Qu.:3.490	3rd Qu.: 0.8200	3rd Qu.:12.50
Max. :1.0998	Max. :6.210	Max. : 4.1800	Max. :25.60
	NA's :104	NA's :310	NA's :185

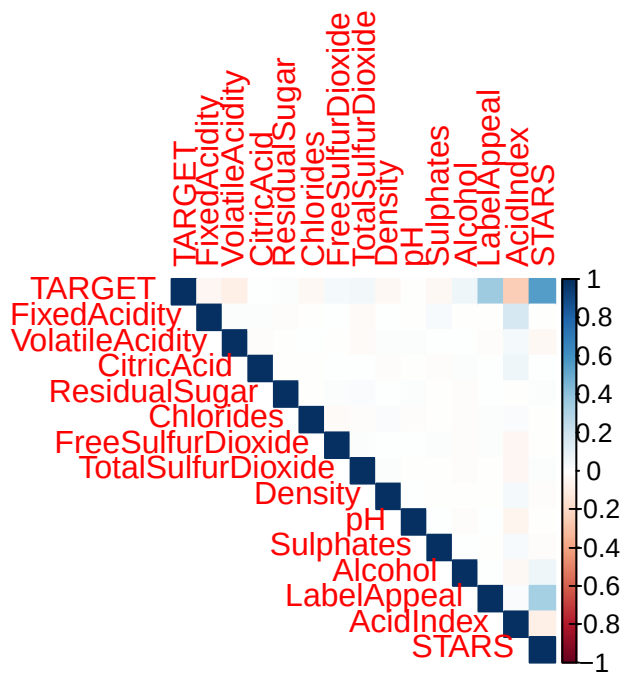
LabelAppeal	AcidIndex	STARS
Min. : -2.00000	Min. : 5.000	Min. :1.00
1st Qu.: -1.00000	1st Qu.: 7.000	1st Qu.:1.00
Median : 0.00000	Median : 8.000	Median :2.00
Mean : 0.01349	Mean : 7.748	Mean :2.04
3rd Qu.: 1.00000	3rd Qu.: 8.000	3rd Qu.:3.00
Max. : 2.00000	Max. :17.000	Max. :4.00
		NA's :841

Correlation Plot

```
library(corrplot)
```

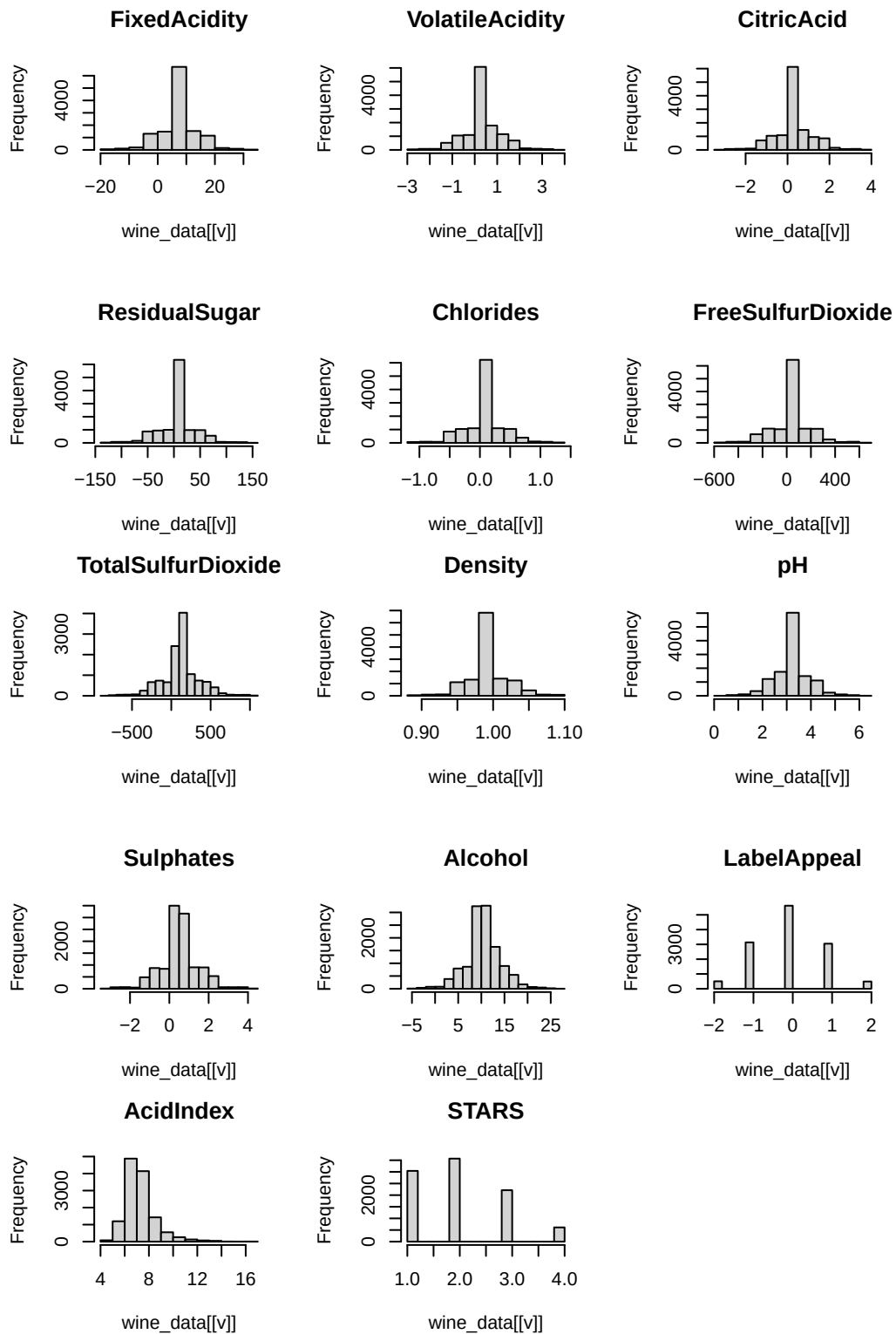
```
corrplot 0.95 loaded
```

```
corr_p<-corrplot(cor_mat, method = "color", type = "upper")
```



Distributions

```
par(mfrow=c(2,3))
for (v in names(wine_data)[names(wine_data) != "TARGET"]) {
  hist(wine_data[[v]], main=v, col="lightgrey")
}
```



Target Distribution

```
hist(wine_data$TARGET)
```



Most variables in the dataset display fairly balanced, centered distributions, consistent with standardized chemical measures. `FixedAcidity`, `VolatileAcidity`, `CitricAcid`, `Density`, `pH`, `Alcohol`, and similar predictors do not exhibit strong skew and therefore do not require transformation.

`AcidIndex` is the main exception, showing a noticeably right-skewed distribution. Because of this, we will bucket `AcidIndex` into categories rather than apply a mathematical transform, as bucketing provides a more stable and interpretable way to handle its nonlinearity.

The `STARS` rating variable is discrete (1–4) with many missing values. Instead of treating missing `STARS` as zeros or excluding them, we will recode missing values as “Not Rated”, so wines without an expert rating are not implicitly penalized or interpreted as having low quality.

Other variables, including `LabelAppeal`, are well-behaved and can be used in their original numeric form. Thus, only `AcidIndex` and `STARS` require special treatment in the preparation step.

The predictor variables also show minimal correlation with one another.

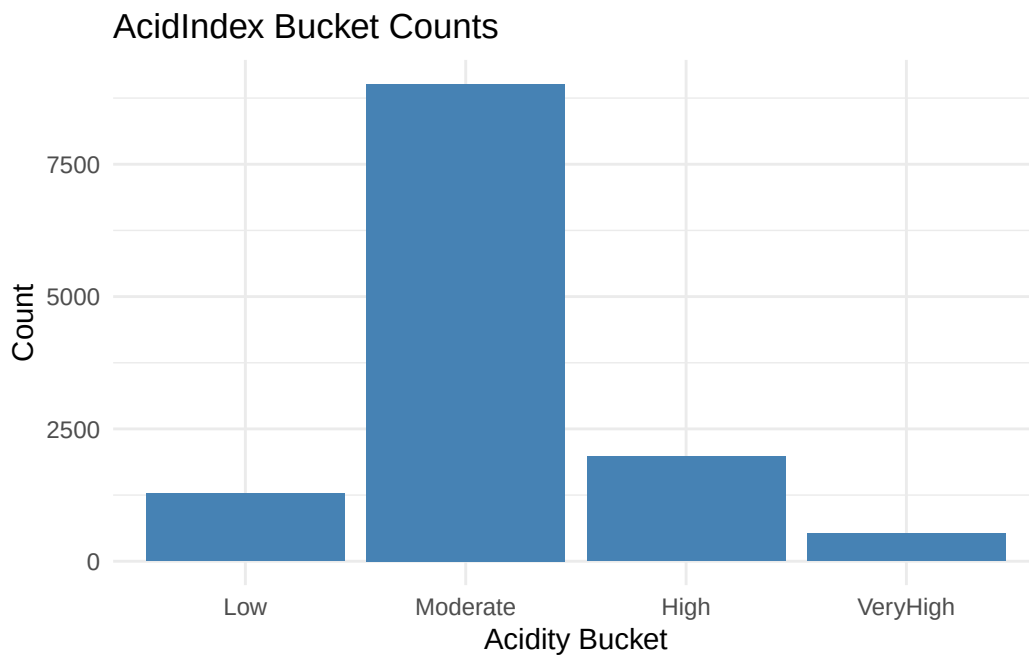
The `TARGET` variable ranges from 0 to 8 and shows a moderately right-skewed distribution. The most common purchase counts fall between 2 and 5 cases, with a noticeable spike at 0 but

no severe zero inflation. Very few wines reach the upper end of the range, and the distribution decreases steadily after 4 cases. Overall, the shape of **TARGET** is consistent with a typical count outcome, supporting the use of Poisson and Negative Binomial regression models.

Data Preparation

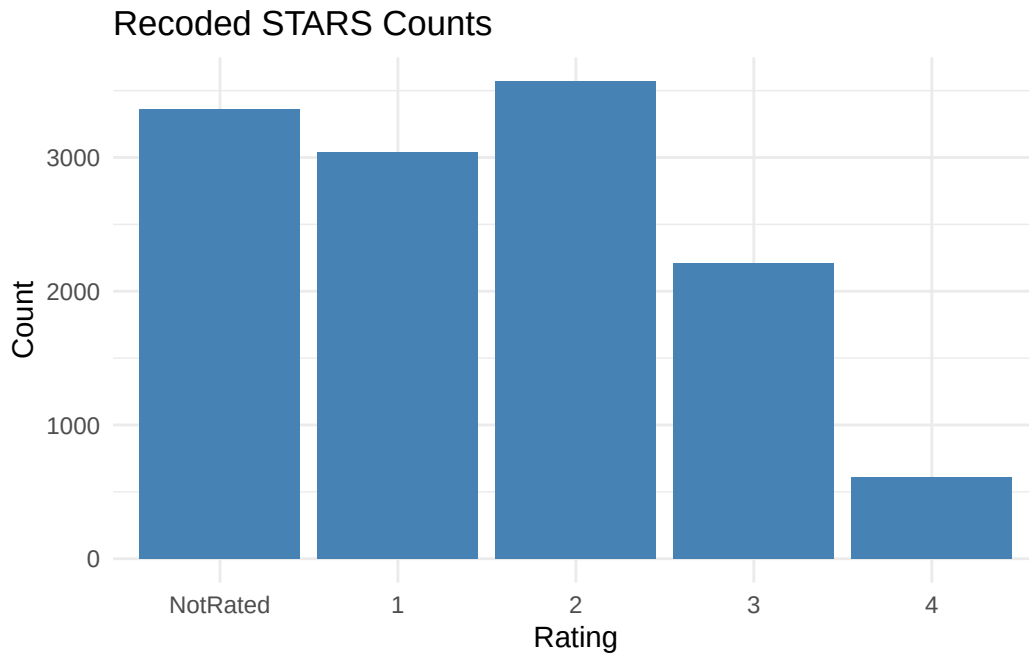
Bucketing AcidIndex

```
acid_bucket_plt
```



STAR

```
STARS_plt
```



NA's

```
# Variables with missing values
vars_with_na <- c(
  "ResidualSugar",
  "Chlorides",
  "FreeSulfurDioxide",
  "TotalSulfurDioxide",
  "Sulphates",
  "Alcohol",
  "pH"
)

# 1. Create missingness flags
for (v in vars_with_na) {
  wine_data[[paste0(v, "_missing")]] <- ifelse(is.na(wine_data[[v]]), 1, 0)
}

# 2. Median imputation
for (v in vars_with_na) {
  med <- median(wine_data[[v]], na.rm = TRUE)
  wine_data[[v]][is.na(wine_data[[v]])] <- med
}
```

```

}
train_medians <- sapply(
  wine_data[, vars_with_na],
  median,
  na.rm = TRUE
)
# 3. Verify no missing values remain
colSums(is.na(wine_data))

```

TARGET	FixedAcidity
0	0
VolatileAcidity	CitricAcid
0	0
ResidualSugar	Chlorides
0	0
FreeSulfurDioxide	TotalSulfurDioxide
0	0
Density	pH
0	0
Sulphates	Alcohol
0	0
LabelAppeal	AcidIndex
0	0
STARS	AcidIndex_bucket
0	0
STARS_missing	ResidualSugar_missing
0	0
Chlorides_missing	FreeSulfurDioxide_missing
0	0
TotalSulfurDioxide_missing	Sulphates_missing
0	0
Alcohol_missing	pH_missing
0	0

Test Mirroring Data Preparation

```

test_acid_plt<-ggplot(wine_test, aes(x = AcidIndex_bucket)) +
  geom_bar(fill = "steelblue") +
  theme_minimal() +
  labs(title = "AcidIndex Buckets in Test Data",

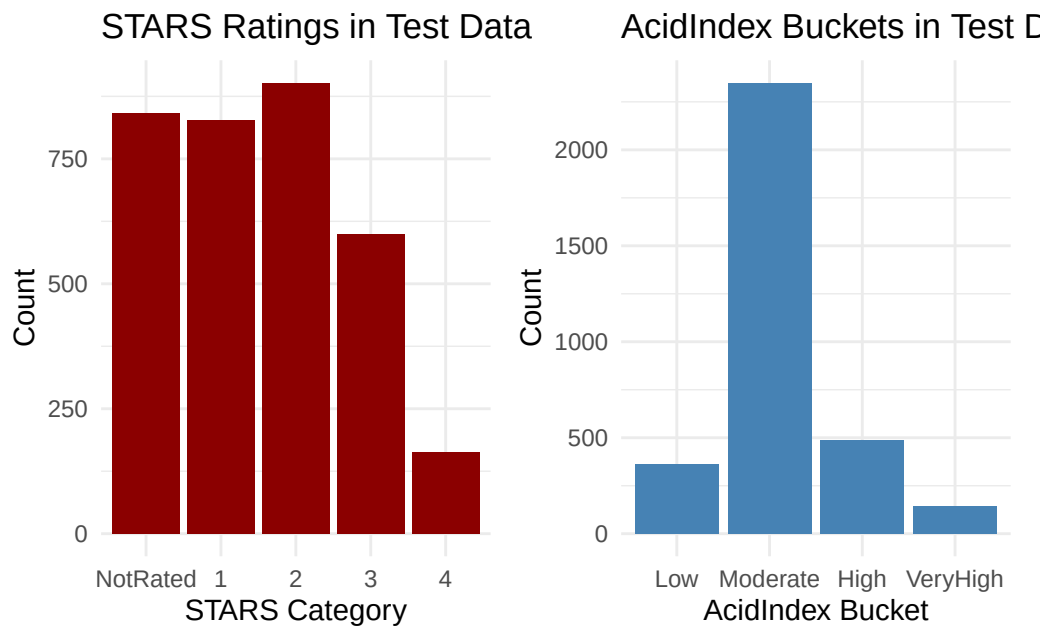
```

```

    x = "AcidIndex Bucket",
    y = "Count")
test_stars_plt<-ggplot(wine_test, aes(x = STARS)) +
  geom_bar(fill = "darkred") +
  theme_minimal() +
  labs(title = "STARS Ratings in Test Data",
       x = "STARS Category",
       y = "Count")

test_stars_plt + test_acid_plt

```



Splitting

```

set.seed(123)

# total number of rows
n <- nrow(wine_data)

# randomly sample 75% of the data for training
train_idx <- sample(seq_len(n), size = 0.75 * n)

# create training and validation sets

```

```
wine_train <- wine_data[train_idx, ]
wine_valid <- wine_data[-train_idx, ]

cat("Training rows:", nrow(wine_train), "\n")
```

Training rows: 9596

```
cat("Validation rows:", nrow(wine_valid), "\n")
```

Validation rows: 3199

Several chemical variables contained missing values, including ResidualSugar, Chlorides, FreeSulfurDioxide, TotalSulfurDioxide, Sulphates, and Alcohol. For each of these, a binary missingness indicator was created, as missing measurements may carry predictive value. The original numeric values were then imputed using the median, which maintains the shape of each distribution while preventing the loss of observations. All data preparation was then replicated on the test set. Finally the training data set was split 75-25 for model development.

Building Models

In this section we will explore different models to predict our target we will begin with a baseline linear model, then a multiple linear regression followed by poisson and count models.

Multiple Linear Model simple

```
### =====
### Linear Regression Model 1 (Baseline)
### =====

lm1 <- lm(TARGET ~ STARS + LabelAppeal, data = wine_train)
summary(lm1)
```

Call:

```
lm(formula = TARGET ~ STARS + LabelAppeal, data = wine_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.6534	-0.8097	0.1606	0.7848	5.8759

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.24784	0.02673	46.68	<2e-16 ***
STARS1	1.45917	0.03863	37.77	<2e-16 ***
STARS2	2.52924	0.03722	67.96	<2e-16 ***
STARS3	3.15337	0.04279	73.69	<2e-16 ***
STARS4	3.85136	0.06803	56.62	<2e-16 ***
LabelAppeal	0.43815	0.01592	27.53	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.332 on 9590 degrees of freedom

Multiple R-squared: 0.5221, Adjusted R-squared: 0.5218

F-statistic: 2095 on 5 and 9590 DF, p-value: < 2.2e-16

```
# Evaluate on validation set
pred_lm1 <- predict(lm1, newdata = wine_valid)

mae_lm1 <- mean(abs(wine_valid$TARGET - pred_lm1))
rmse_lm1 <- sqrt(mean((wine_valid$TARGET - pred_lm1)^2))
add_results("LM1_Baseline", "Linear", lm1, wine_valid, pred_lm1)
cat("LM1 MAE:", mae_lm1, "\n")
```

LM1 MAE: 1.075446

```
cat("LM1 RMSE:", rmse_lm1, "\n")
```

LM1 RMSE: 1.35614

Multiple Linear Regression-full

```
### =====
### Linear Regression Model 2 (Expanded)
### =====

lm2 <- lm(TARGET ~ STARS + LabelAppeal + AcidIndex_bucket + Alcohol,
```

```
data = wine_train)
```

```
summary(lm2)
```

Call:

```
lm(formula = TARGET ~ STARS + LabelAppeal + AcidIndex_bucket +  
    Alcohol, data = wine_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.8482	-0.8858	0.0855	0.7617	5.6638

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.480921	0.063081	23.476	< 2e-16 ***
STARS1	1.382811	0.038017	36.374	< 2e-16 ***
STARS2	2.419329	0.036841	65.669	< 2e-16 ***
STARS3	3.007103	0.042538	70.693	< 2e-16 ***
STARS4	3.705497	0.067032	55.279	< 2e-16 ***
LabelAppeal	0.460043	0.015623	29.446	< 2e-16 ***
AcidIndex_bucketModerate	-0.138504	0.044698	-3.099	0.00195 **
AcidIndex_bucketHigh	-0.574929	0.054135	-10.620	< 2e-16 ***
AcidIndex_bucketVeryHigh	-1.292294	0.078810	-16.397	< 2e-16 ***
Alcohol	0.008279	0.003690	2.244	0.02488 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.304 on 9586 degrees of freedom

Multiple R-squared: 0.542, Adjusted R-squared: 0.5416

F-statistic: 1260 on 9 and 9586 DF, p-value: < 2.2e-16

```
# Evaluate on validation set
```

```
pred_lm2 <- predict(lm2, newdata = wine_valid)
```

```
mae_lm2 <- mean(abs(wine_valid$TARGET - pred_lm2))
```

```
rmse_lm2 <- sqrt(mean((wine_valid$TARGET - pred_lm2)^2))
```

```
add_results("LM2_Expanded", "Linear", lm2, wine_valid, pred_lm2)
```

```
cat("LM2 MAE:", mae_lm2, "\n")
```

LM2 MAE: 1.036195

```
cat("LM2 RMSE:", rmse_lm2, "\n")
```

LM2 RMSE: 1.32258

Poisson regression base

```
pois1 <- glm(  
  TARGET ~ STARS + LabelAppeal ,  
  family = poisson(link = "log"),  
  data = wine_train  
)  
pred_pois1 <- predict(pois1, newdata = wine_valid, type = "response")  
add_results("Poisson Base", "Poisson", pois1, wine_valid, pred_pois1)  
  
# predictions on validation  
  
summary(pois1)
```

Call:

```
glm(formula = TARGET ~ STARS + LabelAppeal, family = poisson(link = "log"),  
     data = wine_train)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.176676	0.018426	9.588	<2e-16 ***
STARS1	0.806472	0.022597	35.690	<2e-16 ***
STARS2	1.143451	0.020982	54.497	<2e-16 ***
STARS3	1.278676	0.021953	58.246	<2e-16 ***
STARS4	1.398505	0.027603	50.664	<2e-16 ***
LabelAppeal	0.150672	0.007007	21.504	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 17109 on 9595 degrees of freedom
Residual deviance: 10456 on 9590 degrees of freedom
AIC: 34464

Number of Fisher Scoring iterations: 6

Poisson Expanded~

```
pois2 <- glm(  
  TARGET ~ .,  
  family = poisson(link = "log"),  
  data = wine_train  
)  
  
pred_pois2 <- predict(pois2, newdata = wine_valid, type = "response")  
  
add_results("Poisson2", "Poisson", pois2, wine_valid, pred_pois2)  
  
summary(pois2)
```

Call:

```
glm(formula = TARGET ~ ., family = poisson(link = "log"), data = wine_train)
```

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	9.134e-01	2.334e-01	3.913	9.11e-05	***
FixedAcidity	-1.072e-04	9.485e-04	-0.113	0.90998	
VolatileAcidity	-3.076e-02	7.530e-03	-4.085	4.41e-05	***
CitricAcid	8.006e-03	6.821e-03	1.174	0.24049	
ResidualSugar	4.275e-05	1.781e-04	0.240	0.81024	
Chlorides	-5.168e-02	1.896e-02	-2.726	0.00641	**
FreeSulfurDioxide	1.069e-04	4.090e-05	2.613	0.00899	**
TotalSulfurDioxide	6.004e-05	2.633e-05	2.280	0.02258	*
Density	-3.835e-01	2.212e-01	-1.734	0.08295	.
pH	-1.295e-02	8.940e-03	-1.449	0.14738	
Sulphates	-8.452e-03	6.599e-03	-1.281	0.20024	
Alcohol	2.331e-03	1.631e-03	1.429	0.15301	
LabelAppeal	1.568e-01	7.030e-03	22.299	< 2e-16	***
AcidIndex	-3.589e-02	1.194e-02	-3.007	0.00264	**
STARS1	7.701e-01	2.267e-02	33.966	< 2e-16	***
STARS2	1.090e+00	2.116e-02	51.512	< 2e-16	***
STARS3	1.212e+00	2.224e-02	54.490	< 2e-16	***
STARS4	1.330e+00	2.789e-02	47.674	< 2e-16	***

AcidIndex_bucketModerate	7.621e-03	2.601e-02	0.293	0.76952
AcidIndex_bucketHigh	-8.445e-02	4.624e-02	-1.827	0.06777 .
AcidIndex_bucketVeryHigh	-4.184e-01	8.700e-02	-4.809	1.52e-06 ***
STARS_missing	NA	NA	NA	NA
ResidualSugar_missing	3.788e-02	2.662e-02	1.423	0.15469
Chlorides_missing	-9.076e-03	2.667e-02	-0.340	0.73362
FreeSulfurDioxide_missing	1.066e-02	2.617e-02	0.407	0.68389
TotalSulfurDioxide_missing	1.304e-02	2.600e-02	0.501	0.61616
Sulphates_missing	-8.413e-03	2.029e-02	-0.415	0.67841
Alcohol_missing	5.783e-03	2.680e-02	0.216	0.82918
pH_missing	-3.480e-02	3.421e-02	-1.017	0.30902

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 17109 on 9595 degrees of freedom
 Residual deviance: 10100 on 9568 degrees of freedom
 AIC: 34152

Number of Fisher Scoring iterations: 6

Negative Binomials simple

```
nb1 <- glm.nb(
  TARGET ~ STARS + LabelAppeal + AcidIndex_bucket + Alcohol,
  data = wine_train
)
```

```
Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
control$trace > : iteration limit reached
Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
control$trace > : iteration limit reached
```

```
pred_nb1 <- predict(nb1, newdata = wine_valid, type = "response")

add_results("NegBin1", "NegBin", nb1, wine_valid, pred_nb1)

summary(nb1)
```

Call:

```
glm.nb(formula = TARGET ~ STARS + LabelAppeal + AcidIndex_bucket +  
        Alcohol, data = wine_train, init.theta = 41085.18864, link = log)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.266190	0.030779	8.649	<2e-16 ***
STARS1	0.775293	0.022656	34.221	<2e-16 ***
STARS2	1.099172	0.021113	52.061	<2e-16 ***
STARS3	1.222140	0.022184	55.090	<2e-16 ***
STARS4	1.342853	0.027801	48.303	<2e-16 ***
LabelAppeal	0.156683	0.007022	22.312	<2e-16 ***
AcidIndex_bucketModerate	-0.044978	0.018646	-2.412	0.0159 *
AcidIndex_bucketHigh	-0.204917	0.024245	-8.452	<2e-16 ***
AcidIndex_bucketVeryHigh	-0.645294	0.047941	-13.460	<2e-16 ***
Alcohol	0.002416	0.001629	1.483	0.1379

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(41085.19) family taken to be 1)

Null deviance: 17108 on 9595 degrees of freedom
Residual deviance: 10158 on 9586 degrees of freedom
AIC: 34176

Number of Fisher Scoring iterations: 1

Theta: 41085

Std. Err.: 39724

Warning while fitting theta: iteration limit reached

2 x log-likelihood: -34154.19

Negative Binomial- Expanded

```
nb2 <- glm.nb(  
  TARGET ~ .,  
  data = wine_train  
)
```

Warning in theta.ml(Y, mu, sum(w), w, limit = control\$maxit, trace =

```
control$trace > : iteration limit reached
Warning in theta.ml(Y, mu, sum(w), w, limit = control$maxit, trace =
control$trace > : iteration limit reached
```

```
pred_nb2 <- predict(nb2, newdata = wine_valid, type = "response")

add_results("NegBin2", "NegBin", nb2, wine_valid, pred_nb2)

summary(nb2)
```

Call:

```
glm.nb(formula = TARGET ~ ., data = wine_train, init.theta = 41353.29552,
link = log)
```

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	z	value	Pr(> z)
(Intercept)	9.134e-01	2.334e-01	3.913	9.11e-05	***
FixedAcidity	-1.072e-04	9.486e-04	-0.113	0.90998	
VolatileAcidity	-3.076e-02	7.530e-03	-4.085	4.41e-05	***
CitricAcid	8.007e-03	6.821e-03	1.174	0.24050	
ResidualSugar	4.276e-05	1.781e-04	0.240	0.81022	
Chlorides	-5.168e-02	1.896e-02	-2.726	0.00641	**
FreeSulfurDioxide	1.069e-04	4.090e-05	2.613	0.00899	**
TotalSulfurDioxide	6.005e-05	2.633e-05	2.280	0.02258	*
Density	-3.835e-01	2.212e-01	-1.734	0.08295	.
pH	-1.295e-02	8.941e-03	-1.449	0.14737	
Sulphates	-8.453e-03	6.599e-03	-1.281	0.20024	
Alcohol	2.331e-03	1.632e-03	1.429	0.15305	
LabelAppeal	1.568e-01	7.031e-03	22.298	< 2e-16	***
AcidIndex	-3.589e-02	1.194e-02	-3.007	0.00264	**
STARS1	7.701e-01	2.267e-02	33.966	< 2e-16	***
STARS2	1.090e+00	2.116e-02	51.511	< 2e-16	***
STARS3	1.212e+00	2.224e-02	54.489	< 2e-16	***
STARS4	1.330e+00	2.789e-02	47.672	< 2e-16	***
AcidIndex_bucketModerate	7.622e-03	2.601e-02	0.293	0.76953	
AcidIndex_bucketHigh	-8.445e-02	4.624e-02	-1.826	0.06778	.
AcidIndex_bucketVeryHigh	-4.184e-01	8.700e-02	-4.809	1.52e-06	***
STARS_missing	NA	NA	NA	NA	
ResidualSugar_missing	3.788e-02	2.662e-02	1.423	0.15471	
Chlorides_missing	-9.076e-03	2.667e-02	-0.340	0.73362	
FreeSulfurDioxide_missing	1.066e-02	2.617e-02	0.407	0.68391	

TotalSulfurDioxide_missing	1.304e-02	2.600e-02	0.501	0.61617
Sulphates_missing	-8.414e-03	2.029e-02	-0.415	0.67840
Alcohol_missing	5.782e-03	2.680e-02	0.216	0.82919
pH_missing	-3.480e-02	3.421e-02	-1.017	0.30902

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(41353.3) family taken to be 1)

Null deviance: 17108 on 9595 degrees of freedom
 Residual deviance: 10100 on 9568 degrees of freedom
 AIC: 34154

Number of Fisher Scoring iterations: 1

Theta: 41353
 Std. Err.: 40018

Warning while fitting theta: iteration limit reached

2 x log-likelihood: -34096.03

Zero Inflated Negative Bionomial

```
# Example ZINB model
zinb1 <- zeroinfl(
  TARGET ~ STARS + LabelAppeal + AcidIndex_bucket + Alcohol |
    STARS + LabelAppeal,      # zero-inflation part
  data = wine_train,
  dist = "negbin"
)
print(zinb1)
```

Call:

```
zeroinfl(formula = TARGET ~ STARS + LabelAppeal + AcidIndex_bucket +
  Alcohol | STARS + LabelAppeal, data = wine_train, dist = "negbin")
```

Count model coefficients (negbin with log link):

(Intercept)	STARS1	STARS2
1.092310	0.068357	0.201405
STARS3	STARS4	LabelAppeal

	0.297727	0.390896	0.231822
AcidIndex_bucketModerate	-0.035173	-0.106168	-0.205610
Alcohol	0.006101		

Theta = 46248909.7777

Zero-inflation model coefficients (binomial with logit link):

(Intercept)	STARS1	STARS2	STARS3	STARS4	LabelAppeal
0.4684	-2.1405	-6.0446	-20.3516	-20.5150	0.7747

```
summary(zinb1)
```

Call:

```
zeroinfl(formula = TARGET ~ STARS + LabelAppeal + AcidIndex_bucket +  
          Alcohol | STARS + LabelAppeal, data = wine_train, dist = "negbin")
```

Pearson residuals:

Min	1Q	Median	3Q	Max
-2.36557	-0.46279	0.01323	0.39424	4.30254

Count model coefficients (negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.092310	0.031759	34.393	< 2e-16 ***
STARS1	0.068357	0.024647	2.773	0.005546 **
STARS2	0.201405	0.022994	8.759	< 2e-16 ***
STARS3	0.297727	0.024053	12.378	< 2e-16 ***
STARS4	0.390896	0.029484	13.258	< 2e-16 ***
LabelAppeal	0.231822	0.007260	31.932	< 2e-16 ***
AcidIndex_bucketModerate	-0.035173	0.018870	-1.864	0.062321 .
AcidIndex_bucketHigh	-0.106168	0.024806	-4.280	1.87e-05 ***
AcidIndex_bucketVeryHigh	-0.205610	0.055296	-3.718	0.000201 ***
Alcohol	0.006101	0.001661	3.673	0.000240 ***
Log(theta)	17.649548	2.513523	7.022	2.19e-12 ***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.46842	0.04698	9.971	<2e-16 ***
STARS1	-2.14047	0.08680	-24.660	<2e-16 ***
STARS2	-6.04460	0.45269	-13.353	<2e-16 ***
STARS3	-20.35157	399.42431	-0.051	0.959

```
STARS4      -20.51499  723.15028  -0.028    0.977
LabelAppeal  0.77465    0.04779  16.210    <2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Theta = 46248909.7777
```

```
Number of iterations in BFGS optimization: 59
```

```
Log-likelihood: -1.544e+04 on 17 Df
```

```
pred_zinb <- predict(zinb1, newdata = wine_valid, type = "response")

# Compute MAE and RMSE for ZINB
mae_zinb  <- mean(abs(wine_valid$TARGET - pred_zinb))
rmse_zinb <- sqrt(mean((wine_valid$TARGET - pred_zinb)^2))

# Extract AIC properly
aic_zinb <- AIC(zinb1)

# Add row to results
new_row <- data.frame(
  Model    = "ZINB1",
  Type     = "ZINB",
  R2       = NA,           # not defined for ZINB
  Adj_R2   = NA,           # not applicable
  MAE      = mae_zinb,
  RMSE     = rmse_zinb,
  AIC      = aic_zinb,
  stringsAsFactors = FALSE
)

model_results <- rbind(model_results, new_row)
```

Selecting Model

Across all models evaluated, the Zero-Inflated Negative Binomial model (ZINB1) demonstrated the strongest overall performance with the **lowest AIC (30,923.52)** and the **lowest MAE and RMSE** among all count-based models. Although the expanded linear regression model achieved competitive error metrics, linear models violate distributional assumptions for discrete count data and cannot serve as the final deployed model. They are retained only as benchmarks for comparison.

Among the traditional count regression models, the Negative Binomial and Poisson variants performed reasonably well, but none matched the predictive accuracy or fit of the ZINB model. Both Poisson models showed higher AIC values and larger prediction errors, reflecting their inability to accommodate overdispersion. The Negative Binomial models improved on this but still failed to capture the substantial probability mass at zero: despite roughly 20% of the training data containing zero purchases, standard Poisson and NB models systematically underpredicted zeros due to the log-link structure, which forces strictly positive expected values.

Introducing a zero-inflation component addressed this deficiency. The ZINB model explicitly separates the data-generating process into (1) a structural zero mechanism and (2) a count process for wines that are actually ordered. This two-part structure yielded superior predictive accuracy and a substantially better AIC than the NB or Poisson models. It also allowed the model to produce realistic zero predictions, something the standard GLM-based count models could not accomplish.

Given its substantially better model fit, stronger predictive accuracy, and more realistic handling of zero outcomes, **ZINB1 was selected as the final model** for generating predictions on the evaluation dataset.

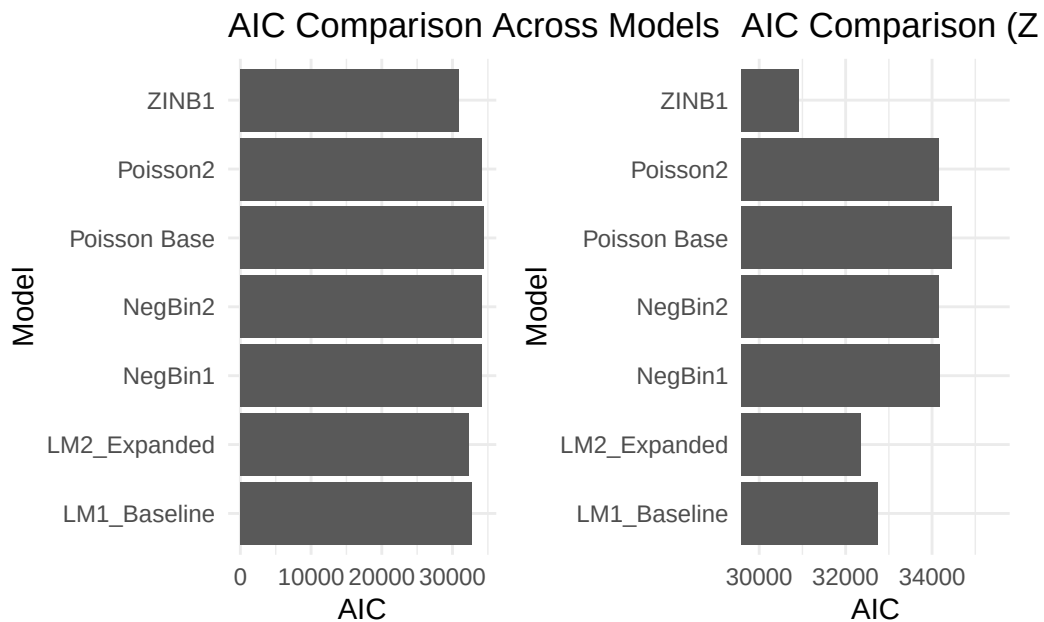
```
model_results|>
  arrange((AIC))
```

	Model	Type	R2	Adj_R2	MAE	RMSE	AIC
1	ZINB1	ZINB	NA	NA	1.010027	1.320891	30923.52
2	LM2_Expanded	Linear	0.5419982	0.5415682	1.036195	1.322580	32343.47
3	LM1_Baseline	Linear	0.5220872	0.5218380	1.075446	1.356140	32743.83
4	Poisson2	Poisson	NA	NA	1.029192	1.313099	34151.71
5	NegBin2	NegBin	NA	NA	1.029193	1.313099	34154.03
6	NegBin1	NegBin	NA	NA	1.024112	1.311981	34176.19
7	Poisson Base	Poisson	NA	NA	1.044851	1.334146	34463.51

```
aic_mod_plt<-ggplot(model_results, aes(x = Model, y = AIC, )) +
  geom_col() +
  coord_flip() +
  theme_minimal() +
  labs(title = "AIC Comparison Across Models",
       x = "Model",
       y = "AIC")
aic_zoom_plt<-ggplot(model_results, aes(x = Model, y = AIC, )) +
  geom_col() +
  coord_flip(xlim = NULL, ylim = c(min(model_results$AIC) - 1050,
                                   max(model_results$AIC) + 1050)) +
```

```
theme_minimal() +
labs(title = "AIC Comparison (Zoomed In)",
     x = "Model",
     y = "AIC")
```

```
aic_mod_plt + aic_zoom_plt
```

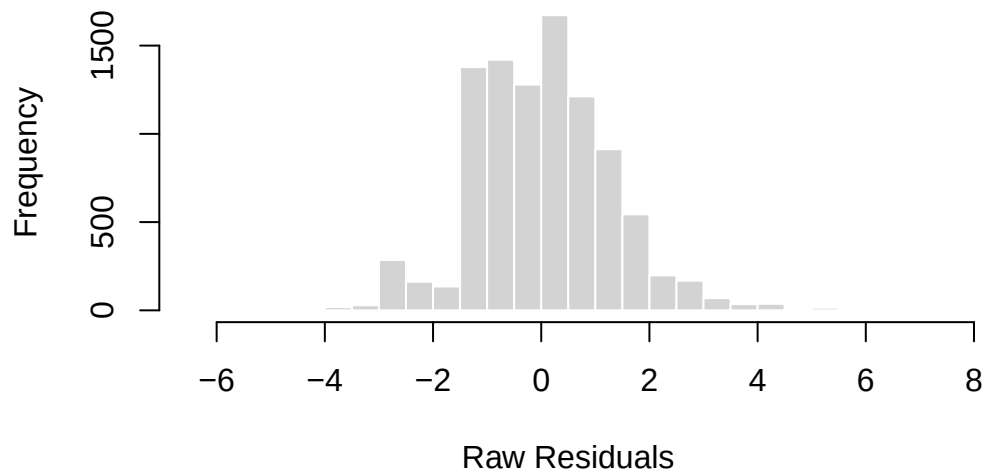


```
# Extract residuals
res_raw      <- residuals(zinb1, type = "response")
res_pearson  <- residuals(zinb1, type = "pearson")

# Extract fitted values
fitted_vals  <- predict(zinb1, type = "response")

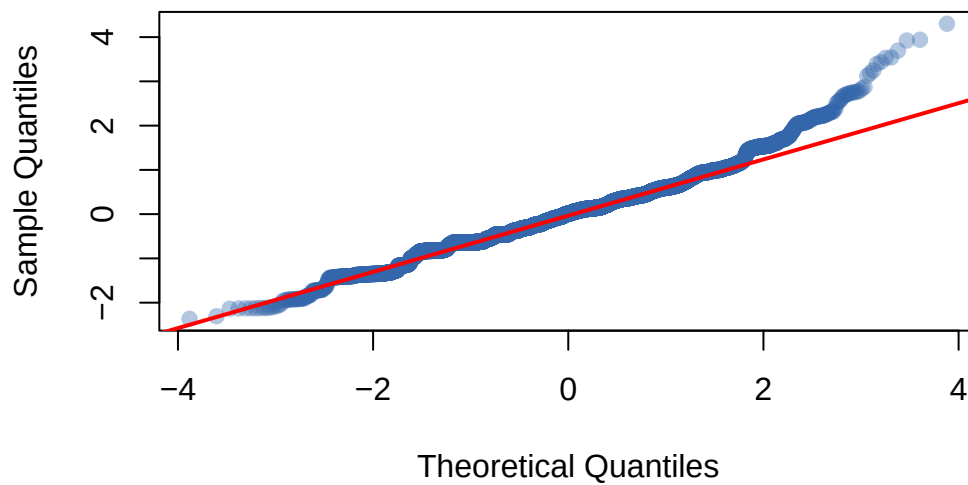
# --- 1. Residual Distribution ---
hist(res_raw,
     breaks = 30,
     col = "lightgrey",
     border = "white",
     main = "ZINB Residual Distribution",
     xlab = "Raw Residuals")
```

ZINB Residual Distribution



```
# --- 2. QQ Plot ---  
qqnorm(res_pearson,  
       main = "ZINB QQ Plot (Pearson Residuals)",  
       pch = 19, col = "#3366AA60")  
qqline(res_pearson, col = "red", lwd = 2)
```

ZINB QQ Plot (Pearson Residuals)

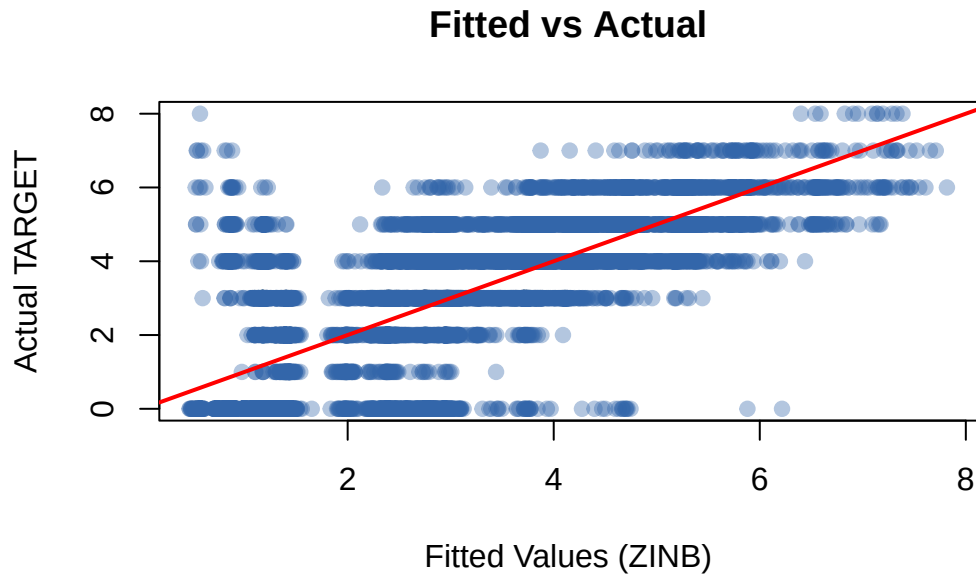


```
# --- 3. Fitted vs Actual ---  
plot(fitted_vals, wine_train$TARGET,  
     pch = 19, col = "#3366AA60",
```

```

xlab = "Fitted Values (ZINB)",
ylab = "Actual TARGET",
main = "Fitted vs Actual")
abline(0, 1, col = "red", lwd = 2)

```



Predictions

```

# 7. Inspect predictions
head(predsdf, 20)

```

	INDEX	PREDICTED_TARGET
1	3	3
2	9	4
3	10	3
4	18	3
5	21	0
6	30	6
7	31	3
8	37	0
9	39	0
10	47	0
11	60	3
12	62	0

13	63	3
14	64	0
15	68	2
16	75	3
17	76	3
18	83	0
19	87	4
20	92	5

```
# 8. Plot prediction distribution
pred_dis <-ggplot(predsdf, aes(x = PREDICTED_TARGET)) +
  geom_histogram(binwidth = 1, fill = "lightgrey", color = "white") +
  labs(title = "Final TARGET Predictions (ZINB)",
       x = "Predicted Case Sales")

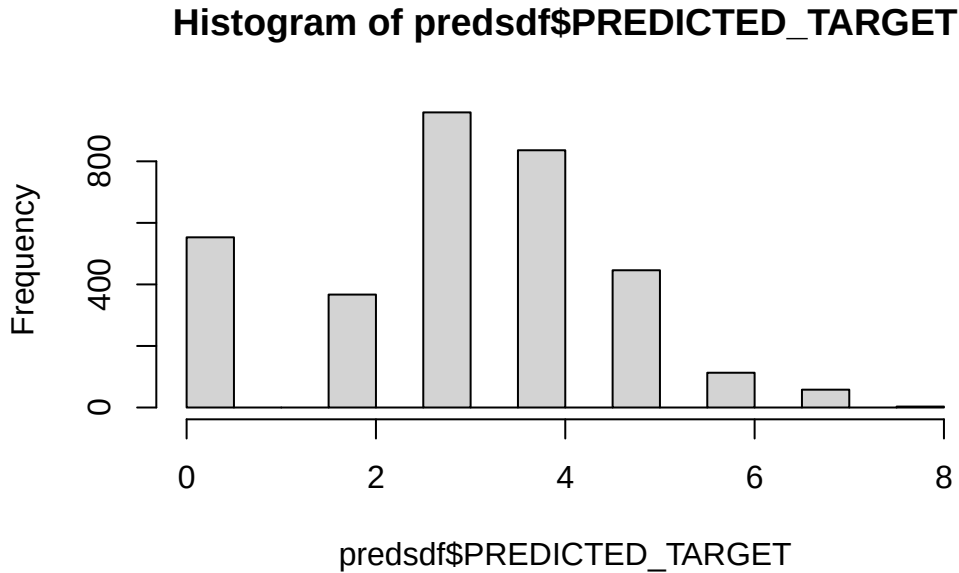
# 9. Save results for submission
write.csv(predsdf, "final_predictions_ZINB.csv", row.names = FALSE)
```

Conclusion

```
hist(wine_train$TARGET)
```



```
hist(predsdf$PREDICTED_TARGET)
```



The comparison between the training **TARGET** distribution and the ZINB model's predicted distribution shows that the model successfully reproduces the overall shape, center, and spread of the observed case-sales behavior. Both histograms exhibit their highest frequencies in the 2–5 case range, indicating that the model accurately captures the most common purchasing patterns seen in the training data.

The model also correctly preserves the long right tail, with non-zero predictions extending up to 7–8 cases, matching the maximum observed values in the training set. Importantly, unlike standard Poisson or Negative Binomial models both of which failed to generate zeros, the ZINB model produces a realistic number of zero predictions driven by its explicit zero-inflation component. Although the height of the zero bar is slightly lower than in the training distribution, the overall magnitude is directionally consistent and represents a substantial improvement over traditional count GLMs.

Across the remaining count values, the predicted distribution aligns closely with the empirical one, with only minor differences in relative frequencies. The slight smoothing around counts 3–5 reflects the expected behavior of a probabilistic regression model producing mean-based predictions rather than exact replications of sample frequencies.

Overall, the ZINB model provides a well-calibrated approximation of the true **TARGET** distribution, capturing both the central mass and the zero-inflation present in the data. This confirms that the model is appropriately tuned for deployment and capable of generating realistic predictions for the evaluation dataset.

Important Predictors

```
print(summary(zinb1))
```

Call:

```
zeroinfl(formula = TARGET ~ STARS + LabelAppeal + AcidIndex_bucket +  
          Alcohol | STARS + LabelAppeal, data = wine_train, dist = "negbin")
```

Pearson residuals:

	Min	1Q	Median	3Q	Max
	-2.36557	-0.46279	0.01323	0.39424	4.30254

Count model coefficients (negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.092310	0.031759	34.393	< 2e-16	***
STARS1	0.068357	0.024647	2.773	0.005546	**
STARS2	0.201405	0.022994	8.759	< 2e-16	***
STARS3	0.297727	0.024053	12.378	< 2e-16	***
STARS4	0.390896	0.029484	13.258	< 2e-16	***
LabelAppeal	0.231822	0.007260	31.932	< 2e-16	***
AcidIndex_bucketModerate	-0.035173	0.018870	-1.864	0.062321	.
AcidIndex_bucketHigh	-0.106168	0.024806	-4.280	1.87e-05	***
AcidIndex_bucketVeryHigh	-0.205610	0.055296	-3.718	0.000201	***
Alcohol	0.006101	0.001661	3.673	0.000240	***
Log(theta)	17.649548	2.513523	7.022	2.19e-12	***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.46842	0.04698	9.971	<2e-16	***
STARS1	-2.14047	0.08680	-24.660	<2e-16	***
STARS2	-6.04460	0.45269	-13.353	<2e-16	***
STARS3	-20.35157	399.42431	-0.051	0.959	
STARS4	-20.51499	723.15028	-0.028	0.977	
LabelAppeal	0.77465	0.04779	16.210	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta = 46248909.7777

Number of iterations in BFGS optimization: 59

Log-likelihood: -1.544e+04 on 17 Df

In the **count model**, **STARS** ratings and **LabelAppeal** are the strongest positive predictors of case sales. Higher **STARS** levels significantly increase expected purchases, and **LabelAppeal** has a large positive effect, confirming that expert ratings and marketing appeal drive higher order volumes. **Alcohol** content also has a small positive effect. In contrast, wines in the **High** and **Very High AcidIndex** buckets show significantly lower expected sales, indicating that high acidity reduces demand.

In the **zero-inflation model**, **STARS** ratings greatly reduce the likelihood of a structural zero, meaning expert-rated wines are far less likely to receive no orders at all. **LabelAppeal** also significantly decreases the chance of a zero outcome. Together, these results show that both expert ratings and label strength influence not only how many cases a wine sells, but also whether it is ordered in the first place.